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Detecting an electrical fault within a monitored zone of a power transmission line

Abstract

Examples for protecting a power transmission line in response to a fault occurring within a monitored zone in a power transmission system are described. In an example, an occurrence of the fault in at least one phase of the power transmission line may be identified. Thereafter, an actual rate of change of incremental current is calculated based on calculated incremental currents. With the actual rate of change determined, a threshold for rate of change of incremental current is calculated based on calculated incremental voltages, the calculated incremental currents, line parameters, and a zone setting for the monitored zone. Based on comparison of the actual rate of change and the threshold for the rate of change, the fault is determined to have occurred in the monitored zone. Thereafter, a trip signal may be generated for controlling a switching device associated with the power transmission line.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4203142	12/1979	Lee	361/87	G01R 27/18
5453903	12/1994	Chow	N/A	N/A
5847913	12/1997	Turner	361/115	G01R 31/52
6625550	12/2002	Scott	361/87	H01H 71/125
6947273	12/2004	Bassett	361/111	G06F 1/26
8749933	12/2013	Berggren	361/65	H02H 7/268
11660978	12/2022	Douglass	361/104	B60L 3/04
2004/0066593	12/2003	Kolker	361/42	G01R 31/52
2004/0263183	12/2003	Naidu	324/536	G01R 31/52
2009/0021878	12/2008	Saha	361/81	G01R 31/52
2011/0013324	12/2010	Hyde	361/54	H02H 7/26
2012/0004869	12/2011	Saarinen	702/59	G01R 31/3274
2013/0107405	12/2012	Blumschein et al.	N/A	N/A
2015/0162741	12/2014	Valdes	361/87	H02H 7/262
2016/0077149	12/2015	Schweitzer, III	307/131	G01R 31/08
2016/0149389	12/2015	Schroeder	361/93.2	H02H 1/0092
2019/0004103	12/2018	Liu	N/A	G01R 31/52

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1262802	12/1999	CN	N/A
102195275	12/2010	CN	N/A

102842890	12/2011	CN	N/A
106199329	12/2015	CN	N/A
107576840	12/2017	CN	N/A
109066608	12/2017	CN	N/A
0577228	12/1993	EP	N/A
H0614447	12/1993	JP	N/A
2002311076	12/2001	JP	N/A
2011078292	12/2010	JP	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion of the International Searching Authority for the corresponding PCT Application No. PCT/EP2021/055071 dated May 31, 2021, 13 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a national stage entry of International Patent App. No. PCT/EP2021/055071, filed on Mar. 1, 2021, which claims priority to Indian Patent Application number 202041008792, filed on Mar. 2, 2020, which are both hereby incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) The present subject matter relates, in general, to power transmission systems. More specifically, the present subject matter relates to approaches for detection of faults within a monitored zone of a power transmission line.

BACKGROUND

(3) Short circuit faults occurring in the transmission line is one of the hazardous phenomena in power systems. Such faults, when they occur, have to be detected and cleared as soon as possible. If the faults are not cleared or addressed within the critical clearing time, the faults may cause the power transmission system to lose transient stability, which in turn may lead to a blackout. A distance relay coupled to an Intelligent Electronic Device (IED) may be utilized to provide protection of transmission lines against such faults. As would be understood, the IED may monitor the impedance during its operation. When the impedance monitored by the IED is less than predefined threshold the distance relay may be activated to ensure protection against the fault. Since the distance relays act in response to the detection of a fault, it may be understood, that speed of the conventional distance relay depends on the accuracy of phasor estimation during transmission line fault conditions.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The features, aspects, and advantages of the present subject matter will be better understood with regards to the following description and accompanying figures. The use of the same reference number in different figures indicate similar or identical features and components.

- (2) FIG. 1 provides an illustration depicting an electrical network having an intelligent electronic device, as per one example;
- (3) FIG. 2 provides a block diagram of an example intelligent electronic device, as per one example;
- (4) FIGS. 3-7 provides illustrative graphs depicting occurrence of a three-phase fault within a monitored zone of a power transmission line, the instantaneous values of current and voltage, and corresponding incremental current and incremental voltage, as per one example; and
- (5) FIG. 8 provides a flow diagram depicting an example method for ascertaining the presence of a fault within a monitored zone of a power transmission line within an electrical network.
- (6) It may be noted that throughout the drawings, identical reference numbers designate similar, but not necessarily identical, elements. The figures are not necessarily to scale, and the size of some parts may be exaggerated to more clearly illustrate the example shown.
- (7) Moreover, the drawings provide examples and/or implementations consistent with the description; however, the description is not limited to the examples and/or implementations provided in the drawings.

DETAILED DESCRIPTION

(8) Electrical fault may be considered as deviation of voltages and/or currents values which may occur due to external or internal changes in an electrical network. Under normal operating conditions, electrical network equipment in transmission lines carry normal voltages and currents and operate within their normal operating parameters. However, during occurrence of electrical faults, as a result of which excessively high currents may flow through such network equipment. This may cause damage to equipment and devices within the electrical network. Many preventive measures are conventionally implemented to safeguard the electrical network. In one such example, when the impedance drops below a predefined value, an IED may cause a distance relay to detect the fault and accordingly generates a trip command for a switching device, such as a circuit breaker, to trip to prevent any damages that may result from the impedance drops experienced in the electrical network. Faults may be broadly categorized asymmetric and symmetric faults. Asymmetrical fault is a condition in which load in a three-phase power supply becomes unequal on all the three phases of the transmission line. On the other hand, symmetrical faults or balanced faults may be considered as such faults which impact each of the three-phases of the electrical network equally and at the same time. Examples of such symmetric faults include, but are not limited to, line-to-line-to-line (L-L-L), line-to-line-to line-to ground (L-L-L-G).

(9) One of the techniques employed for detection of a fault and/or protection of the transmission line from these faults, involves an approach which is commonly known as the distance protection method. In such cases, a short circuit on a transmission line may be detected based on measurement of short-circuit impedance. To this end, the protection devices, such as an IED, may measure the impedance up until the fault location. Based on this, a further determination may be made to ascertain whether the fault in fact occurs within the transmission line, or a zone, which is to be protected by the protection device. If it is ascertained that the short circuit that has occurred lies on the line to be protected, the line in question is disconnected and the faulty network portion is separated from the system.

(10) As would also be understood, power transmission may include power generation from one or more sources. Presently, power generation may also involve renewable source of power which may tend to reduce inertia and transient stability margin of the electrical network. Owing to the effect on the stability, high-speed protection is necessary. Conventional techniques for high-speed protection are based on certain time domain protection principles. Such technique may further employ high sampling rates and processing power for high-speed line protection. Such approaches generally have been able to reduce fault detection times, depending on the fault location and source-to-line impedance ratio. However, considering that the involvements of such renewable sources within the electrical network will only increase, approaches for quicker detection of faults is therefore desired.

(11) Approaches for detecting occurrence of electrical faults which may have occurred within a monitored zone of a power transmission line, are described. In an example, an occurrence of the fault in at least one phase of the power transmission line may be initially identified. Continuing further, an actual rate of change of incremental current and incremental voltage based on calculated incremental currents and voltages is determined. The incremental currents are such that they correspond to the difference in currents. In a similar way, incremental voltages may be considered as values which correspond to difference in voltages. The currents and the voltages, in the present example, are measured at the terminals of the power transmission line.

(12) With the actual rate of change of incremental current determined, a threshold for the rate of change of incremental current may be further determined. The threshold for the rate of change of the incremental current may be understood as the rate of change of such incremental current assuming that a fault has occurred at the zone boundary. The threshold for the rate of change of the incremental current then forms the basis for ascertaining whether the fault under consideration is occurred in the zone boundary or not. In an example, the threshold for the rate of change of incremental current may be calculated based on the calculated incremental voltages, the calculated incremental currents, line parameters, and a zone setting for the monitored zone. It may be noted that other parameters may also be utilized for calculating the threshold for the rate of change of incremental current. Zone setting (reach setting) is the setting which determines the reach of protection, i.e. the zone expected to be covered by the protection, and beyond which the protection will not operate. This can be in terms of percentage of transmission line length or impedance.

(13) Thereafter, a further determination may be made to ascertain whether the fault may have occurred within a monitored zone. A monitored zone may be considered as a portion of the power transmission line which is monitored by an Intelligent Electronic Device (IED). In an example, the determination may be based on a comparison of the actual rate of change of the incremental current and the threshold for the rate of change of the incremental current. For example, the actual rate of change of the incremental current being greater than the threshold of the rate of change of the incremental current is indicative of a fault. On the other hand, if the actual rate of change of incremental current is less than the threshold for the rate of change of incremental current, it may be concluded that the fault may have occurred beyond the monitored zone. On determining that the fault has occurred within the monitored zone, a trip signal for controlling a switching device, such as a circuit breaker, associated with the power transmission line may be generated.

(14) The approaches as described also enable determining whether fault occurs within reach and should be rectified within less time as compared to conventional approaches, thereby preventing any adverse impact of faults, for example, such as those which arise due to current transformer saturation. It has also been observed that a power transmission system implementing the present subject matter is able to detect and remain stable in situations of occurrence of faults during power swing conditions. The manner in which the above-mentioned steps may be differ across various example, without any deviation to the scope of protection which has been sought. These and other aspects are further described in detail in conjunction with the accompanying FIGS. 1-5.

(15) FIG. 1 provides a block diagram of an equivalent electrical network **100**, as per an example. The electrical network **100** comprises transmission line **102** and two-electrical sources, namely sources **104**, **106**. The transmission line **102** is further provided with one or more switching device(s) **108-1, 2, 3, 4, . . . , N** (collectively referred to as switching device(s) **108**). The switching device(s) **108** allow opening of the circuit to restrict the flow of excess current in the electrical network during fault condition. It may be noted that the electrical network **100** as depicted is only illustrative. The electrical network **100** may include further components without deviating from the scope of the present subject matter.

(16) The electrical network **100** is further installed with an intelligent electronic device **110** (referred to as the IED **110**). The IED **110** may be in electrical communication with the transmission lines **102** either directly or through other connecting means. The IED **110**, during

operation, may receive and monitor measurements from one or more measurement devices.

Examples of such measurement devices include, but are not limited to, current transformers and voltage transformers. Based on the measurements data received, the IED **110** may generate one or more signals to control the switching device(s) **108**, as explained in the following paragraphs.

(17) The IED **110** further includes a phase selection module **112** and fault detection module **114**. The phase selection module **112** and fault detection module **114** may be implemented as either software installed within the IED **110**, or as hardware in the form of electronic circuitry integrated within the circuitry of the IED **110**. The present example is described considering that the electrical network **100** is experiencing an electrical fault. The present subject matter is capable of detecting occurrence of such faults within a monitored zone, when the IED **110** is placed at terminal A within the transmission line **102** such that current signals from the measurement devices may be obtained. It may be noted that the IED **110** may also be adapted to determine the occurrence of a power swing without deviating from the scope of the present subject matter.

(18) In operation, the phase selection module **112** determines occurrence of fault within a particular phase of the transmission lines **102**. The manner in which the phase, in which the fault has occurred, may be determined is further explained in conjunction with other figures. Once the phase in which the fault has occurred is determined, the IED **110** may further determine whether the fault has occurred within the monitored zone of the transmission lines **102**. In an example, the fault detection module **114** may determine whether the fault has occurred within the monitored zone.

(19) To such an end, the fault detection module **114** may determine values of currents and voltages at the terminal of the transmission lines **102**. Based on the current values and the voltage values, the fault detection module **114** may further determine incremental currents based on the changes in the currents measured at the terminals. In a similar manner, the incremental voltages, based on the changes in the voltages may also be determined.

(20) Once the above-mentioned incremental values are determined, the fault detection module **114** may further calculate an actual rate of change of incremental current (referred to as the actual rate of change, for sake of brevity) based on the determined incremental currents. Thereafter, the fault detection module **114** may further calculate a threshold for the rate of change of the incremental current (referred to as threshold for the rate of change, for sake of brevity). In an example, the threshold for the rate of change of incremental current may be calculated based on the calculated incremental voltages, the calculated incremental currents, and line parameters (e.g., resistance or inductance) corresponding to the transmission lines **102** present within the monitored zone.

(21) As also described previously, the threshold for the rate of change may be considered as the rate of change of incremental current which would be present assuming that a fault is to have occurred at the zone boundary. In an example, the zone boundary may be a length which is a portion of the length of the transmission lines **102**. In another example, the zone boundary may be defined as occurring at about 80% of the length of the transmission lines **102**. It may be noted that the present example defining the length at which the zone boundary is present is only indicative. Any other measure defining where the zone boundary is located may be used without deviating from the scope of the present subject matter.

(22) Returning to the present example, once the actual rate of change and the threshold for the rate of change of the incremental currents are obtained, the fault detection module **114** may further process the same to obtain a processed actual rate of change and a processed threshold for the rate of change of the incremental currents. In an example, the fault detection module **114** may calculate a root-mean-square values of the actual rate of change and the threshold for the rate of change to provide the processed actual rate of change and the processed threshold for the rate of change.

(23) The fault detection module **114** may then subsequently assess whether the fault has occurred within the zone boundary based on the processed actual rate of change and the processed threshold for the rate of change. For example, the fault detection module **114** may compare the processed actual rate of change and the processed threshold for the rate of change. If the processed actual rate

of change is less than the processed threshold for the rate of change, the fault detection module **114** may indicate that the fault has occurred beyond the monitored zone boundary. However, if the processed actual rate of change is greater than the processed threshold rate of change, the fault detection module **114** may accordingly indicate that the fault has occurred within the monitored zone boundary.

(24) On determining the occurrence of fault within the monitored zone boundary, the fault detection module **114** may further generate one or more trip signals for a switching device(s) **108**, e.g., a circuit breaker, which may be coupled to the transmission lines **102** under consideration. Based on the trip signals, the switching device(s) **108** may get activated to isolate the fault which has occurred within the monitored zone boundary. These and other examples are further described in conjunction with FIG. 2.

(25) FIG. 2 provides a block diagram of an intelligent electronic device (IED) **110**, as per one example. The IED **110** includes processor(s) **202**, interface(s) **204** and a memory(s) **206**. The processor(s) **202** may be a single processing unit or may include a number of units, all of which could include multiple computing units. The processor(s) **202** may be implemented as one or more microprocessor, microcomputers, digital signal processors, central processing units, state machines, logic circuitries, and/or any devices that manipulate signals based on operational instructions. Among other capabilities the processor(s) **202** are adapted to fetch and execute processor-readable instructions stored in the memory **206** to implement one or more functionalities.

(26) The interface(s) **204** may include a variety of software and hardware enabled interfaces. The interface(s) **204** may enable the communication and connectivity between the IED **110** and other components of the electrical network **100**. Examples of such components include, but is not limited to, switching device(s) **108** and sensors. The interface(s) **204** may facilitate multiple communications within a wide variety of protocols and may also enable communication with one or more computer enabled terminals or similar network components.

(27) The memory(s) **206** may be coupled to the processor(s) **202**. The memory(s) **206** may include any computer-readable medium known in the art including, for example, volatile memory, such as Static Random-Access Memory (SRAM) and Dynamic Random-Access Memory (DRAM), and/or non-volatile memory, such as Read Only Memory (ROM), Erasable Programmable ROMs (EPROMs), flash memories, hard disks, optical disks, and magnetic tapes.

(28) The IED **110** may further include one or more module(s) **208**. The module(s) **208** may be implemented as a combination of hardware and programming (for example, programmable instructions) to implement a variety of functionalities of the module(s) **208**. In examples described herein, such combinations of hardware and programming may be implemented in several different ways. For example, the programming for the module(s) **208** may be executable instructions. Such instructions in turn may be stored on a non-transitory machine-readable storage medium which may be coupled either directly with the IED **110** or indirectly (for example, through networked means). In example implemented as a hardware, the module(s) **208** may include a processing resource (for example, either a single processor or a combination of multiple processors), to execute such instructions. In the present examples, the processor-readable storage medium may store instructions that, when executed by the processing resource, implement module(s) **208**. In other examples, module(s) **208** may be implemented by electronic circuitry.

(29) The data **212** includes data that is either stored or generated as a result of functionalities implemented by any of the module(s) **208**. It may be further noted that information stored and available in data **212** may be utilized for detecting the zone within which the fault is to have occurred. In an example, the module(s) **208** include phase selection module **112**, fault detection module **114** and other module(s) **210**. The other module(s) **210** may implement functionalities that supplement applications or functions performed by the IED **110** or any of the module(s) **208**. The data **212** on the other hand may include pre-fault loop voltage **214**, a pre-fault loop current **216**, an incremental voltage **218**, an incremental current **220**, an actual rate of change of incremental

current **222**, a threshold for the rate of change of incremental current **224**, a processed rate of change of incremental current **226**, a processed threshold for the rate of change of incremental current **228** and other data **230**. In addition, the IED **110** may further include other component(s) **232**. Such other component(s) **232** may include a variety of other electrical components that enable functionalities of managing and controlling the operation of the electrical network **100**. Examples of such other component(s) **232** include, but is not limited to, relays, controllers, switches and voltage regulators.

(30) The IED **110** detects occurrence of an electrical fault within a monitored zone of a power transmission line within the electrical network, such as electrical network **100**. The operation of the IED **110** is further described in conjunction with FIGS. 3-4. FIG. 3-4 provides a series of illustrative graphs depicting the electric current waveforms. It may be noted that the waveforms thus depicted are only indicative and may be pertinent to the present example. The waveforms may differ slightly depending on the implementation.

(31) Returning to the present example, the IED **110** may be connected with one or more measurement devices installed within the electrical network **100** through interface(s) **204**. As mentioned previously, examples of such measurement devices include a current transformer or a voltage transformer. In operation, the phase selection module **112** may monitor the current and voltage measurements corresponding to the three-phase electric current that is being transmitted within the electrical network **100**, to identify the phase in which a fault has occurred.

(32) In an example, while monitoring a flow of electric current within the electrical network **100**, a phase selection module **112** is to determine a state of a start-up signal. Based on the state of the start-up signal the phase selection step may be initiated. In an example, the phase selection module **112** may process the three-phase input currents. Further, the phase selection module **112** may determine phase-phase quantities by calculating a moving average of the input currents for each phase. Based on the moving average of the input currents, the phase selection module **112** may determine one or more phase-phase quantities based on which the start-up signal may be generated. The moving average and the phase-phase quantities may be determined based on the equations 1-6, as indicated below:

$$(33) \quad i_{Ast}^a(k) = \frac{1}{N} \cdot \text{Math}_{i=(k-N+1)}^k i_A^a(i) \quad (1) \quad i_{Ast}^b(k) = \frac{1}{N} \cdot \text{Math}_{i=(k-N+1)}^k i_A^b(i) \quad (2)$$

$$i_{Ast}^c(k) = \frac{1}{N} \cdot \text{Math}_{i=(k-N+1)}^k i_A^c(i) \quad (3) \quad i_{Ast}^{ab}(k) = i_{Ast}^a(k) - i_{Ast}^b(k) \quad (4)$$

$$i_{Ast}^{bc}(k) = i_{Ast}^b(k) - i_{Ast}^c(k) \quad (5) \quad i_{Ast}^{ca}(k) = i_{Ast}^c(k) - i_{Ast}^a(k) \quad (6) \quad \text{where, } i.\text{sub.Ast.sup.a};$$

$i.\text{sub.Ast.sup.b}$, $i.\text{sub.Ast.sup.c}$, $i.\text{sub.Ast.sup.ab}$, $i.\text{sub.Ast.sup.bc}$ and $i.\text{sub.Ast.sup.ca}$ are the instantaneous samples of currents; k is the current sample; and N is the number of samples per power cycle.

(34) In one example, the magnitude of instantaneous samples of currents $i.\text{sub.Ast.sup.a}$, $i.\text{sub.Ast.sup.b}$, $i.\text{sub.Ast.sup.c}$, $i.\text{sub.Ast.sup.ab}$, $i.\text{sub.Ast.sup.bc}$ and $i.\text{sub.Ast.sup.ca}$ are compared against a threshold $i.\text{sup.th}$. In one example, the threshold $i.\text{sup.th}$ may be 0.01. If the phase selection module **112** determines any of the currents $i.\text{sub.Ast.sup.a}$, $i.\text{sub.Ast.sup.b}$, $i.\text{sub.Ast.sup.c}$, $i.\text{sub.Ast.sup.ab}$, $i.\text{sub.Ast.sup.bc}$ and $i.\text{sub.Ast.sup.ca}$ to be greater than a threshold $i.\text{sup.th}$ a phase-selection start-up signal may be generated.

(35) Once the phase-selection start-up signal is generated, the fault detection module **114** may further calculate a value of the incremental currents. In an example, the incremental quantity of an electrical signal at any instant is defined as, the difference between instantaneous magnitude of the signal at an instant and the magnitude of the same signal at such an instance in a previous power cycle. In one example, incremental quantity is calculated for phase currents as well as phase-phase currents, which is based on the following equations:

$$\Delta i.\text{sub.A.sup.a}(k) = i.\text{sub.A.sup.a}(k) - i.\text{sub.A.sup.a}(k-N) \quad (7)$$

$$\Delta i.\text{sub.A.sup.b}(k) = i.\text{sub.A.sup.b}(k) - i.\text{sub.A.sup.b}(k-N) \quad (8)$$

$$\Delta i_{\text{sub.A.sup.c}}(k) = \Delta i_{\text{sub.A.sup.c}}(k) - \Delta i_{\text{sub.A.sup.c}}(k-N) \quad (9)$$

$$\Delta i_{\text{sub.A.sup.ab}}(k) = \Delta i_{\text{sub.A.sup.a}}(k) - \Delta i_{\text{sub.A.sup.b}}(k) \quad (10)$$

$$\Delta i_{\text{sub.A.sup.bc}}(k) = \Delta i_{\text{sub.A.sup.b}}(k) - \Delta i_{\text{sub.A.sup.c}}(k) \quad (11)$$

$$\Delta i_{\text{sub.A.sup.ca}}(k) = \Delta i_{\text{sub.A.sup.c}}(k) - \Delta i_{\text{sub.A.sup.a}}(k) \quad (12)$$

Where, $i_{\text{sub.A.sup.p}}(k)$ is k .sup.th sample of the secondary current signal at terminal A, and where p stands for the corresponding phase, $p \in (a, b, c)$;

$i_{\text{sub.A.sup.p}}(k-N)$ is $(k-N)$.sup.th sample of the current signal measured at terminal A, $p \in (a, b, c)$;

N is the number of samples in one power cycle;

$\Delta i_{\text{sub.A.sup.p}}(k)$ is the k .sup.th sample of incremental quantity of phase current, $p \in (a, b, c)$; and

$\Delta i_{\text{sub.A.sup.pp}}(k)$ is the k .sup.th sample of incremental quantity of phase current, $pp \in (ab, bc, ca)$

(36) Once the incremental quantities of different current signals, i.e., incremental currents have been obtained, the phase selection module **112** may further process the same. In an example, the phase selection module **112** may obtain incremental currents using a moving average filter. In one example, the pre-processing is obtained using one or more passes. In first pass, the incremental quantities is passed through a moving average filter of one power cycle window to obtain a first absolute value, $\Delta i_{\text{sub.Ama1.sup.p}}(k)$, which is based upon equation:

$$(37) \quad i_{\text{Ama1}}^p(k) = \frac{1}{N} \cdot \text{Math.}_{j=k-N+1}^k i_A^p(j) \cdot \text{Math.}, p \in (a, b, c, ab, bc, ca) \quad (13)$$

Where, N is the number of samples in one power cycle

$\Delta i_{\text{sub.Ama1.sup.p}}(k)$ is k .sup.th sample of output of first moving average filter, $p \in (a, b, c, ab, bc, ca)$

(38) In second pass, the output of the first moving average filter may be passed through a second moving average filter of half power cycle window and is to obtain second absolute value,

$\Delta i_{\text{sub.Ama2.sup.p}}(k)$ which is based upon equation:

$$(39) \quad i_{\text{Ama2}}^p(k) = \frac{1}{N/2} \cdot \text{Math.}_{j=k-N/2+1}^k i_{\text{Ama1}}^p(j) \cdot \text{Math.}, \quad (14)$$

$p \in (a, b, c, ab, bc, ca)$

Where, N is the number of samples in one power cycle

$\Delta i_{\text{sub.Ama2.sup.p}}(k)$ is k .sup.th sample of output of second moving average filter, $p \in (a, b, c, ab, bc, ca)$

(40) Once the processing has been performed, the phase selection module **112** identifies a fault loop, such that maximum and minimum among phase and phase-phase quantities are identified for each sample. In one example, the maximum and minimum among phase and phase-phase quantities identified for each sample, is based on following equations:

(41) For maximum and minimum current values among phase quantities:

$$\Delta i_{\text{sub.max.sup.pg}}(k) = \max(\Delta i_{\text{sub.Ama2.sup.a}}(k), \Delta i_{\text{sub.Ama2.sup.b}}(k), \Delta i_{\text{sub.Ama2.sup.c}}(k)) \quad (15)$$

$$\Delta i_{\text{sub.min.sup.pg}}(k) = \min(\Delta i_{\text{sub.Ama2.sup.a}}(k), \Delta i_{\text{sub.Ama2.sup.b}}(k), \Delta i_{\text{sub.Ama2.sup.c}}(k)) \quad (16)$$

For maximum and minimum current values among phase-phase quantities:

$$\Delta i_{\text{sub.max.sup.pp}}(k) = \min(\Delta i_{\text{sub.Ama2.sup.ab}}(k), \Delta i_{\text{sub.Ama2.sup.bc}}(k), \Delta i_{\text{sub.Ama2.sup.ca}}(k)) \quad (17)$$

$$\Delta i_{\text{sub.min.sup.pp}}(k) = \min(\Delta i_{\text{sub.Ama2.sup.ab}}(k), \Delta i_{\text{sub.Ama2.sup.bc}}(k), \Delta i_{\text{sub.Ama2.sup.ca}}(k)) \quad (18)$$

(42) In one example, the fault within the loop may correspond to a single-phase fault or phase-phase (i.e., three phase) fault. For determining type of fault, a comparison may be made between the determined absolute value and a pre-determined value (i.e., E which in one example is set as 0.001). To such an end, the phase selection module **112** may define a counter value. In one example, the magnitude of instantaneous samples of currents (i.e. $|\Delta i_{\text{sub.Ama2.sup.ab}}(k)|$, |

$|\Delta i.\text{sub.Ama2.sup.bc}(k)|$, $|\Delta i.\text{sub.Ama2.sup.ca}(k)|$) for each sample is compared with a predefined parameter E . Based on the comparison, the phase selection module **112** may increment the counter value. Once the counter value exceeds a predefined threshold, either a single-phase fault or a three-phase fault may be identified.

(43) The above approach is described in conjunction with the following example. In the present example, six counters are defined corresponding to each phase or a phase-phase loops. For example, Counter A, B and C may correspond to phases a, b and c. Further exemplary Counters AB, BC, CA may be defined, wherein such respective counters correspond to ab, bc, and ca phases. The manner in which each of the Counters A, B, C, AB, BC, CA are incremented may be determined based on the following equations as explained below:

(44) Increment counter A if:

$$\Delta i.\text{sub.max.sup.pg}(k) = |\Delta i.\text{sub.Ama2.sup.a}(k)| \quad (19)$$

$$\Delta i.\text{sub.min.sup.pp}(k) = |\Delta i.\text{sub.Ama2.sup.bc}(k)| \quad (20)$$

$$|\Delta i.\text{sub.Ama2.sup.bc}(k)| < \epsilon \quad (21)$$

Increment Counter B if:

$$\Delta i.\text{sub.max.sup.pg}(k) = |\Delta i.\text{sub.Ama2.sup.b}(k)| \quad (22)$$

$$\Delta i.\text{sub.min.sup.pp}(k) = |\Delta i.\text{sub.Ama2.sup.ca}(k)| \quad (23)$$

$$|\Delta i.\text{sub.Ama2.sup.ab}(k)| < \epsilon \quad (24)$$

Increment Counter C if:

$$\Delta i.\text{sub.max.sup.pg}(k) = |\Delta i.\text{sub.Ama2.sup.c}(k)| \quad ((25_)$$

$$\Delta i.\text{sub.min.sup.pp}(k) = |\Delta i.\text{sub.Ama2.sup.ab}(k)| \quad (26)$$

$$|\Delta i.\text{sub.Ama2.sup.ab}(k)| < \epsilon \quad (27)$$

Increment Counter AB if:

$$\Delta i.\text{sub.min.sup.pg}(k) = |\Delta i.\text{sub.Ama2.sup.c}(k)| \quad (28)$$

$$\Delta i.\text{sub.max.sup.pp}(k) = |\Delta i.\text{sub.Ama2.sup.ab}(k)| \quad (29)$$

Increment Counter BC if:

$$\Delta i.\text{sub.min.sup.pg}(k) = |\Delta i.\text{sub.Ama2.sup.c}(k)| \quad (30)$$

$$\Delta i.\text{sub.max.sup.pp}(k) = |\Delta i.\text{sub.Ama2.sup.ab}(k)| \quad (31)$$

Increment Counter CA if:

$$\Delta i.\text{sub.min.sup.pg}(k) = |\Delta i.\text{sub.Ama2.sup.c}(k)| \quad (32)$$

$$\Delta i.\text{sub.max.sup.pp}(k) = |\Delta i.\text{sub.Ama2.sup.ab}(k)| \quad (33)$$

(45) If each of the $|\Delta i.\text{sub.Ama2.sup.ab}(k)|$, $|\Delta i.\text{sub.Ama2.sup.bc}(k)|$, $|\Delta i.\text{sub.Ama2.sup.ca}(k)|$) is less than ϵ , the phase selection module **112** may determine occurrence of a single phase fault.

Thereafter, the phase selection module **112** may further determine to check if any of the Counters A, B, or C may have reached a predefined counter limit, to determine in which phase, i.e. phases a, b, c, has the fault occurred. For example, if Counter A reaches a predefined counter limit (say, 10), the phase selection module **112** may determine that a single-phase fault has occurred in phase a. In a similar manner, the occurrence of single-phase fault, i.e., a line fault, may be determined by the phase selection module **112** in either phases b or c depending on the counter which reaches the predefined counter limit, and thereafter, may generate a trip signal for the corresponding phase.

(46) If each of the $|\Delta i.\text{sub.Ama2.sup.ab}(k)|$, $|\Delta i.\text{sub.Ama2.sup.bc}(k)|$, $|\Delta i.\text{sub.Ama2.sup.ca}(k)|$) is greater than ϵ , the phase selection module **112** may determine occurrence of a phase-phase fault or a three-phase fault. For example, in case any of the Counters AB, BC, or CA reaches the predefined counter limit, the phase selection module **112** may conclude that a phase-phase fault or a three-phase fault has occurred in the corresponding phase-phase loop and accordingly may generate a trip signal. It may be noted that the above example is only illustrative and is not the only process to identify phases in which the faults have occurred. Other examples for detecting the occurrence of a fault may also be utilized without deviating from the scope of the present subject matter.

(47) Post detecting the occurrence of a fault, the fault detection module **114** may ascertain whether the fault in the transmission lines **102** has occurred within the monitored zone boundary or not. In

an example, the fault detection module **114** receives a measurement of the pre-fault loop voltage **214** and a pre-fault loop current **216**. The fault detection module **114** may further obtain currents and voltages, by measuring the same at the terminals of the transmission lines **102**. In an example, the waveforms **302** and **304**, depicting the measured voltages and currents, respectively, in one example, as depicted in FIGS. 3A-3B. Further, the fault detection module **114** also receives a measurement of an incremental voltage **218** and incremental current **220**, wherein the incremental voltage **218** and the incremental current **220** correspond to difference in voltage and current, respectively, measured during the fault and/or measured prior to occurrence of the fault, which is based on the following equation:

$$\Delta i_{\text{sub.k}} = (i_{\text{sub.k}} - i_{\text{sub.k-N}}) \quad (34)$$

$$\Delta v_{\text{sub.k}} = (v_{\text{sub.k}} - v_{\text{sub.k-N}}) \quad (35)$$

wherein: $\Delta i_{\text{sub.k}}$ is the incremental current; $\Delta v_{\text{sub.k}}$ is the incremental voltage; k is the current sample; and N is the number of samples in one cycle.

It may be noted that the above equation is only illustrative with the first set being determined through other mechanisms as well.

(48) Waveforms corresponding to the incremental voltage **218** and the incremental current **220** are further depicted in FIGS. 3C-3D, respectively. FIGS. 3C-3D are illustrative graphs depicting the incremental voltage **218** and incremental current **220**, as waveforms **306**, **308**. It may also be noted that the depicted graphs are only indicative, and corresponds to one of the many other examples, which also fall within the scope of the present subject matter. Once the incremental voltage **218** and the incremental current **220** are determined, the fault detection module **114** also processes the pre-fault loop voltage **214**, the pre-fault loop current **216**, the incremental voltage **218**, and the incremental current **220** in order to reduce noise to provide processed values for pre-fault loop voltage, pre-fault loop current, incremental voltage and incremental current. The pre-processing performed by the fault detection module **114** may be performed for smoothening the samples in order to isolate errors that may have been caused due to noise or any other unwanted components.

(49) In an example, the pre-processing may be performed by the fault detection module **114** based on the following equations:

$$(50) \quad v_{\text{lpmma}}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \quad v_{\text{lp}}(k-i) \quad (36) \quad i_{\text{lpma}}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \quad i_{\text{lp}}(k-i) \quad (37)$$

$$i_{\text{lprma}}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \quad i_{\text{lpr}}(k-i) \quad (38) \quad v_{\text{ma}}^{\text{pre}}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \quad v^{\text{pre}}(k-i) \quad (39)$$

$$i_{\text{ma}}^{\text{pre}}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \quad i^{\text{pre}}(k-i), \quad (40)$$

(51) wherein $\Delta v_{\text{sub.lpmma}}$ being the processed incremental voltage; $\Delta i_{\text{sub.lplma}}$ being the processed incremental current across an inductive component of the transmission line; $\Delta i_{\text{sub.lprma}}$ being the processed incremental current across a resistive component of the transmission line; $v_{\text{sub.ma.sup.pre}}$ being the processed pre-fault loop voltage; and $i_{\text{sub.ma.sup.pre}}$ being the processed pre-fault loop current.

(52) Once the aforementioned values are obtained, the fault detection module **114** calculates an actual rate of change of incremental current **222** based on the processed incremental current. The value of actual rate of change of incremental current **222** may be calculated based on the following equation:

$$(53) \quad d \quad i_{\text{act}}(k) = \frac{(i_{\text{lplma}}(k) - i_{\text{lplma}}(k-1))}{(\Delta t)} \quad (41)$$

wherein, Δt is a time difference in seconds.

(54) In addition to the above, the fault detection module **114** may also calculates the threshold for the rate of change of incremental current **224** based on the processed incremental voltage, processed incremental current and the voltage measured at a boundary (i.e. at boundary A of the transmission line **102**) of the monitored zone prior to the occurrence of the fault. In one example, the threshold for the rate of change of incremental current is calculated based on the following

equation:

$$(55) \quad i_{thr}(k) = \frac{v_{lpma}(k) - R_1 * z_b * i_{lpma}(k) + v_{z1}^{pre}(k)}{L_1 * z_b} \quad (42)$$

wherein $z_{sub.b}$ denotes the boundary of the monitored zone, and

wherein $R_{sub.1}$, $L_{sub.1}$ are positive-sequence series resistance and inductance of the transmission line.

(56) Post calculations, the fault detection module **114** determines the root-mean-square value of the rate of change of the incremental current **222** and the threshold for the rate of change of incremental current **224** to provide a processed rate of change of incremental current **226** and a processed threshold for the rate of change of incremental current **228**, which may be represented as:

$$(57) \quad d i_{act}^{rms} = \sqrt{\frac{\sum_{i=0}^N (d i_{act}(k-i))^2}{N}} \quad (43) \quad d i_{thr}^{rms} = \sqrt{\frac{\sum_{i=0}^{N/3} (d i_{thr}(k-i))^2}{N/3}} \quad (44)$$

wherein $d\Delta i_{sub.act.sup.rms}$ depicts the processed rate of change of the current; and

$d\Delta i_{sub.thr.sup.rms}$ depicts the processed threshold for the rate of change of incremental current.

(58) Thereafter, the fault detection module **114** further determines whether the fault is to have occurred in the monitored zone based on a comparison between the processed rate of change of the incremental current **226** and the processed threshold for the rate of change of incremental current **228**. In one example, when the processed rate of change of the incremental current **226** is greater than the processed threshold for the rate of change of incremental current **228**, the fault detection module **114** identifies occurrence of the fault within the monitored zone. Conversely, when the processed rate of change of the incremental current **226** is lesser than the processed threshold for the rate of change of incremental current **228**, the fault detection module **114** indicates occurrence of the fault outside the monitored zone. The waveforms **402**, **404** pertaining comparison between the processed rate of change of the incremental current **226** and the processed threshold for the rate of change of incremental current **228** is shown in FIG. 4.

(59) In one embodiment, proposed method is tested with special examples such as CT saturation, power swing, faults during power swing, and the like. These and other aspects are further described in detail in conjunction with the accompanying FIGS. 5-7. In one example, considering a fault scenario under CT saturation. FIG. 5 shows the performance of the proposed algorithm for a fault example with CT saturation. The test example has a fault at 2% of the line (4 km for a 200 km line) with low fault resistance of 0.1 ohm. Typically, FIG. 5A shows the terminal current measured at local terminal A as shown in waveform **502**, FIG. 5B shows incremental quantity of the current as shown in waveform **504**, and FIG. 5C made comparison of actual rate of change of current with threshold as shown in waveforms **506** and **508** respectively. It can be seen from both the waveforms that the fault current is as high as 10 kA and CT has saturated (encircled portions of the waveforms as depicted in FIG. 5B).

(60) In another example, fault detection within a zone which may occur during a power swing. FIG. 6 shows the corresponding waveforms depicting during power swing. Typically, FIG. 6A shows terminal voltage during power swing as shown in waveform **602**, FIG. 6B shows terminal current during power swing as shown in waveform **604**, FIG. 6C shows incremental current **220** during power swing, and FIG. 6D shows comparison between the actual rate of change of incremental current **222** with threshold for the rate of change of incremental current **224** as shown in waveforms **606** and **608** respectively. As it can be observed from the waveforms, the rate of change of current does not cross the threshold and hence the distance element does not operate. In yet another example, a Power Swing Block function (e.g., as implemented by the IED **110**), blocks distance relay to prevent operation during power swings. However, if a fault occurs during a power swing, distance protection of transmission lines is to operate reliably. FIG. 7 shows the corresponding waveforms depicting faults detected during power swing. Currently, an unblocking function is used to detect faults during power swing and unblock distance relay. The proposed solution as explained in previous section does not require any blocking during power swings. It

works accurately for faults during power swing, as shown in various waveforms of FIG. 7.

(61) For an instance, FIG. 7A shows fault location to be identified within 20 kms on a 200 kms transmission line. In one example, fault inception time to be 5 seconds as shown in waveform **702** and detection time to be approximately 5.002 sec, as shown in waveform **704**. Such an analysis further indicates the fault to be present within the monitored zone. It may be noted that experimental graph of FIG. 7A was captured within 2 ms after occurrence of the fault. FIG. 7B shows fault location to be identified within 100 kms on a 200 kms transmission line. In one example, fault inception time to be 5 seconds as shown in waveform **706** and detection time to be approximately 5.003 sec, as shown in waveform **708**. Such an analysis also indicates the fault to be present within the monitored zone. It may be noted that experimental graph of FIG. 7B was captured within 3 ms after occurrence of the fault.

(62) FIG. 7C shows fault location to be identified within 128 kms on a 200 kms transmission line. In one example, fault inception time to be 5 seconds as shown in waveform **710** and detection time to be approximately 5.003 sec, as shown in waveform **712**. Such an analysis also indicates the fault to be present within the monitored zone. It may be noted that experimental graph of FIG. 7C was captured within 3 ms after occurrence of the fault. FIG. 7D shows fault location to be identified within 128 kms on a 200 kms transmission line. In one example, fault inception time to be 5 seconds as shown in waveform **714** and detection time to be almost negligible, as shown in waveform **716**. Such an analysis indicates the fault to be present outside the monitored zone. Hence, this solution can potentially eliminate the need for power swing blocking or unblocking of zone-1 distance relay.

(63) FIG. 8 illustrates a flowchart of a method **800** for detecting occurrence of electrical fault within a monitored zone of a power transmission line in a power transmission system, in accordance with one implementation of the present subject matter. The order in which the methods are described is not intended to be construed as a limitation, and any number of the described method blocks may be combined in any order to implement the methods, or an alternative method. Furthermore, method **800** may be implemented by processing resource through any suitable hardware, non-transitory machine-readable instructions, or combination thereof.

(64) At block **802**, the IED **110** receives a measurement of a pre-fault loop voltage **214** and a pre-fault loop current **216**. In one example, the measurements of the pre-fault loop voltage **214** and the pre-fault loop current **216** may correspond to initial values of current and voltage obtained from one end of the power transmission line.

(65) At block **804**, the IED **110** identifies an occurrence of fault within a particular phase. In one example the loop identification in a particular phase may correspond to each phase of one of a three-phase electric current or a three-phase electric voltage. The occurrence of the fault in a loop may be identified based upon pre-determined conditions, as expressed earlier in equations (19)-(33). Upon detection of fault, a trip signal may be generated.

(66) At block **806**, the IED **110** determines a measurement of an incremental voltage **218** and an incremental current **220**. In one example, the incremental voltage **218** and the incremental current **220** may correspond to difference in voltage and current, respectively, measured during the fault and/or measured prior to occurrence of the fault. An equation based on such a determination is expressed earlier in equations (34)-(35).

(67) At block **808**, the IED **110** calculates an actual rate of change of incremental current **222** based on a processed incremental current. The processed incremental current is based on the incremental current **220**. In one example, the processed incremental current is measured at a boundary of the monitored zone prior to the occurrence of the fault. An equation based on which the processed incremental current and the actual rate of change of incremental current **222** is determined is expressed earlier in equations (36)-(41).

(68) At block **810**, the IED **110** further calculates a threshold for the rate of change of incremental current **224** based on the processed incremental voltage. In one example, the processed incremental

voltage is measured at a boundary of the monitored zone prior to the occurrence of the fault. An equation based on which the threshold for the rate of change of incremental current **224** is determined, is expressed earlier in equation (42)

(69) At block **812**, the IED **110** further determines whether the fault is to have occurred in the monitored zone based on a comparison between a processed rate of change of incremental current **226** and the processed threshold for the rate of change of incremental current **228**. In one example, if the processed rate of change of the incremental current **226** is greater than the processed threshold for the rate of change of incremental current **228**, then the fault detection module **114** indicates occurrence of the fault within the monitored zone. Conversely, when the processed rate of change of the incremental current **226** is lesser than the processed threshold for the rate of change of incremental current **228**, the fault detection module **114** identifies occurrence of the fault outside the monitored zone. The equations based on which the processed rate of change of incremental current **226** and the processed threshold for the rate of change of incremental current **228** is determined, is expressed as equations (43)-(44).

(70) Although implementations of present subject matter have been described in language specific to structural features and/or methods, it is to be noted that the present subject matter is not necessarily limited to the specific features or methods described. Rather, the specific features and methods are disclosed and explained in the context of a few implementations for the present subject matter

Claims

1. A method for protecting a power transmission line in response to a fault within a monitored zone in a power transmission system, the method comprising: identifying an occurrence of a fault in at least one phase of a power transmission line; calculating an actual rate of change of incremental current based on calculated incremental currents; calculating a threshold for rate of change of incremental current based on calculated incremental voltages, the calculated incremental currents, line parameters, and a zone setting for the monitored zone; determining the fault to be in the monitored zone based on comparison of the actual rate of change of incremental current and the threshold for the rate of change of incremental current; and generating a trip signal for controlling a switching device associated with the power transmission line, based on the comparison.
2. The method as claimed in claim 1, wherein the calculated incremental voltages and the calculated incremental currents correspond to difference in voltages and currents, respectively, wherein the voltages and currents are measured at terminals of the power transmission line.
3. The method as claimed in claim 2, wherein calculating the actual rate of change of incremental current further comprises: processing, based on a moving average filter, the calculated incremental currents to provide processed incremental currents; and calculating the actual rate of change of incremental current based on the processed incremental currents.
4. The method as claimed in claim 2, wherein identifying the occurrence of the fault in at least one phase is based on the calculated incremental currents associated with one terminal of the power transmission line.
5. The method as claimed in claim 1, wherein the line parameters comprise resistance and inductance of the transmission line.
6. The method as claimed in claim 1, wherein determining the fault further comprises: calculating a root-mean-square value of the actual rate of change of the incremental current and the threshold for the rate of change of incremental current to provide a processed actual rate of change of incremental current and a processed threshold for the rate of change of incremental current; and comparing a processed actual rate of change of incremental current and a processed threshold for the rate of change of incremental current.
7. The method as claimed in claim 6, wherein the fault is to have occurred when the processed

actual rate of change of incremental current to be greater than the processed threshold for the rate of change of incremental current.

8. A method for protecting a power transmission line in response to a fault within a monitored zone in a power transmission system, the method comprising: identifying an occurrence of a fault in at least one phase of a power transmission line; calculating an actual rate of change of incremental current based on calculated incremental currents, wherein calculating the actual rate of change of incremental current comprises processing, based on a moving average filter, the calculated incremental currents to provide processed incremental currents, and calculating the actual rate of change of incremental current based on the processed incremental currents; calculating a threshold for rate of change of incremental current based on calculated incremental voltages, the calculated incremental currents, line parameters, and a zone setting for the monitored zone; determining the fault to be in the monitored zone based on comparison of the actual rate of change of incremental current and the threshold for the rate of change of incremental current; and generating a trip signal for controlling a switching device associated with the power transmission line, based on the comparison, wherein a processed incremental voltage and the processed incremental currents are

determined using the following equations: $v_{lpma}(k) = \frac{1}{3} \cdot \text{Math.}_{i=-1}^1 v_{lp}(k-i)$

$i_{lpma}(k) = \frac{1}{3} \cdot \text{Math.}_{i=-1}^1 i_{lp}(k-i)$ $i_{lprma}(k) = \frac{1}{3} \cdot \text{Math.}_{i=-1}^1 i_{lpr}(k-i)$ $v_{ma}^{pre}(k) = \frac{1}{3} \cdot \text{Math.}_{i=-1}^1 v_{ma}^{pre}(k-i)$
 $i_{ma}^{pre}(k) = \frac{1}{3} \cdot \text{Math.}_{i=-1}^1 i_{ma}^{pre}(k-i)$ wherein, $\Delta v_{sub.lpma}$ is the processed incremental voltage;

$\Delta i_{sub.lpma}$ is the processed incremental current across an inductive component of the transmission line; $\Delta i_{sub.lprma}$ is the processed incremental current across a resistive component of the transmission line; $v_{sub.ma.sup.pre}$ is a processed pre-fault loop voltage; and $i_{sub.ma.sup.pre}$ is a processed pre-fault loop current.

9. An intelligent electronic device (IED) to protect a power transmission line from a fault which is to have occurred in at least one phase of the power transmission line within a monitored zone in a power transmission system, with the IED being provided at one end of the power transmission line connecting at least two terminals with one or more power transmission line, the IED comprising: a microprocessor; an output interface; a phase selection module to: detect occurrence of a fault in at least one phase of a power transmission line; a fault detection module to: calculate incremental voltages and incremental currents, based on voltages and currents measured at terminals of the power transmission line; calculate an actual rate of change of incremental current based on calculated incremental currents; calculate a threshold for the rate of change of incremental current based on calculated incremental voltages, the calculated incremental currents, line parameters, and a zone setting for a monitored zone; compare the actual rate of change of incremental current and the threshold for the rate of change of incremental current to determine the fault to be in the monitored zone; and generate a trip signal for controlling a switching device associated with the power transmission line, based on the comparison.

10. The device as claimed in claim 9, wherein calculating the actual rate of change of incremental current comprises processing, based on a moving average filter, the incremental currents to provide processed incremental currents.

11. An intelligent electronic (IED) to protect a power transmission line from a fault which is to have occurred in at least one phase of the power transmission line within a monitored zone in a power transmission system, with the IED being provided at one end of the power transmission line connecting at least two terminals with one or more power transmission line, the IED comprising: a microprocessor; an output interface; a phase selection module to: detect occurrence of a fault in at least one phase of a power transmission line; a fault detection module to: calculate incremental voltages and incremental currents, based on voltages and currents measured at terminals of the power transmission line; calculate an actual rate of change of incremental current based on calculated incremental currents, wherein calculating the actual rate of change of incremental

current comprises processing, based on a moving average filter, the incremental currents to provide processed incremental currents; calculate a threshold for the rate of change of incremental current based on calculated incremental voltages, the calculated incremental currents, line parameters, and a zone setting for a monitored zone; compare the actual rate of change of incremental current and the threshold for the rate of change of incremental current to determine the fault to be in the monitored zone; and generate a trip signal for controlling a switching device associated with the power transmission line, based on the comparison, wherein a processed incremental voltage and a processed incremental current are determined using the following equations:

$$v_{lpma}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \cdot v_{lp}(k-i) \quad i_{lplma}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \cdot i_{lpl}(k-i)$$

$$i_{lprma}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \cdot i_{lpr}(k-i) \quad v_{ma}^{pre}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \cdot v_{ma}^{pre}(k-i) \quad i_{ma}^{pre}(k) = \frac{1}{3} \cdot \text{Math}_{i=-1}^1 \cdot i_{ma}^{pre}(k-i)$$

wherein, $\Delta v_{sub.lpma}$ is the processed incremental voltage; and $\Delta i_{sub.lplma}$ is the processed incremental current across an inductive component of the transmission line; and $\Delta i_{sub.lprma}$ is the processed incremental current across a resistive component of the transmission line; and $v_{sub.ma.sup.pre}$ is a processed pre-fault loop voltage; and $i_{sub.ma.sup.pre}$ is a processed pre-fault loop current.

12. The device as claimed in claim 11, wherein the fault detection module is to calculate the actual rate of change of incremental current using the following equation:

$$d \quad i_{act}(k) = \frac{(i_{lplma}(k) - i_{lplma}(k-1))}{(\Delta t)} \quad \text{wherein, } \Delta i_{sub.lpkma} \text{ is the processed incremental current across an inductive component of the transmission line; and } \Delta t \text{ is the interval between two consecutive samples.}$$

13. The device as claimed in claim 11, wherein the fault detection module is to calculate the threshold for the rate of change of incremental current using the equation:

$$d \quad i_{thr}(k) = \frac{v_{lpma}(k) - R_1 * z_b * i_{lprma}(k) + v_{z1}^{pre}(k)}{L_1 * z_b} \quad \text{wherein, } \Delta v_{sub.lpma} \text{ is the processed incremental voltage; } \Delta i_{sub.lprma} \text{ is the processed incremental current across a resistive component of the transmission line; } z_{sub.b} \text{ denotes the boundary of the monitored zone; } R_{sub.1}, L_{sub.1} \text{ are positive-sequence series resistance and inductance of the transmission line; and } v_{sub.z1.sup.pre} \text{ is further determined using the following equation:}$$

$$v_{z1}^{pre}(k) = v_{lpma}(k) - R_1 * z_b * i_{lprma}(k) - L_1 * z_b * \frac{(i_{lplma}(k) - i_{lplma}(k-1))}{(\Delta t)}.$$

14. The device as claimed in claim 13, wherein the processed rate of change of incremental current and the processed threshold for the rate of change of incremental current are represented as:

$$d \quad i_{act}^{rms} = \sqrt{\frac{\text{Math}_{i=0}^N (d \quad i_{act}(k-i))^2}{N}}; \text{ and } d \quad i_{thr}^{rms} = \sqrt{\frac{\text{Math}_{i=0}^{N/3} (d \quad i_{thr}(k-i))^2}{N/3}} \quad \text{wherein } d\Delta i_{sub.act.sup.rms}$$

depicts the processed actual rate of change of incremental current; and wherein $d\Delta i_{sub.thr.sup.rms}$ depicts the processed threshold for the rate of change of incremental current.

15. The device as claimed in claim 14, wherein the fault detection module determines an occurrence of the fault to be within the monitored zone when the processed rate of change of incremental current is greater than the processed threshold for the rate of change of incremental current.

16. A non-transitory computer-readable medium having instructions stored thereon, wherein the instructions when executed by a processor, cause the processor to: identify an occurrence of a fault in at least one phase of a power transmission line; calculate an actual rate of change of incremental current based on calculated incremental currents; calculate a threshold for rate of change of incremental current based on calculated incremental voltages, the calculated incremental currents, line parameters, and a zone setting for a monitored zone in a power transmission system; determine the fault to be in the monitored zone based on comparison of the actual rate of change of incremental current and the threshold for the rate of change of incremental current; and generate a trip signal for controlling a switching device associated with the power transmission line, based on

the comparison.

17. The non-transitory computer-readable medium as claimed in claim 16, wherein the calculated incremental voltages and the calculated incremental currents correspond to difference in voltages and currents, respectively, wherein the voltages and currents are measured at terminals of the power transmission line.

18. The non-transitory computer-readable medium as claimed in claim 17, wherein calculating the actual rate of change of incremental current further comprises: processing, based on a moving average filter, the calculated incremental currents to provide processed incremental currents; and calculating the actual rate of change of incremental current based on the processed incremental currents.

19. The non-transitory computer-readable medium as claimed in claim 16, wherein the line parameters comprise resistance and inductance of the transmission line.

20. A non-transitory computer-readable medium having instructions stored thereon, wherein the instructions when executed by a processor, cause the processor to: identify an occurrence of a fault in at least one phase of a power transmission line; calculate an actual rate of change of incremental current based on calculated incremental currents, wherein calculating the actual rate of change of incremental current comprises processing, based on a moving average filter, the calculated incremental currents to provide processed incremental currents, and calculating the actual rate of change of incremental current based on the processed incremental currents; calculate a threshold for rate of change of incremental current based on calculated incremental voltages, the calculated incremental currents, line parameters, and a zone setting for a monitored zone in a power transmission system; determine the fault to be in the monitored zone based on comparison of the actual rate of change of incremental current and the threshold for the rate of change of incremental current; and generate a trip signal for controlling a switching device associated with the power transmission line, based on the comparison, wherein a processed incremental voltage and a processed incremental current is determined using the following equations:

$$\begin{aligned} v_{lpma}(k) &= \frac{1}{3} \cdot \text{Math.} \sum_{i=-1}^1 v_{lp}(k-i) & i_{lpma}(k) &= \frac{1}{3} \cdot \text{Math.} \sum_{i=-1}^1 i_{lpl}(k-i) \\ i_{lprma}(k) &= \frac{1}{3} \cdot \text{Math.} \sum_{i=-1}^1 i_{lpr}(k-i) & v_{ma}^{pre}(k) &= \frac{1}{3} \cdot \text{Math.} \sum_{i=-1}^1 v^{pre}(k-i) & i_{ma}^{pre}(k) &= \frac{1}{3} \cdot \text{Math.} \sum_{i=-1}^1 i^{pre}(k-i) \end{aligned}$$

wherein, $\Delta v_{sub.lmpa}$ is the processed incremental voltage; $\Delta i_{sub.plma}$ is the processed incremental current across an inductive component of the transmission line; $\Delta i_{sub.lprma}$ is the processed incremental current across a resistive component of the transmission line;

$v_{sub.ma.sup.pre}$ is a processed pre-fault loop voltage; and $i_{sub.ma.sup.pre}$ is a processed pre-fault loop current.
